Probabilistic Methods
Lecture Notes

June 3, 2026

Contents

Chapter 1

Events and Probability (Sample Spaces)

This chapter introduces the fundamental concepts of probability theory, including sample spaces, events, and the calculation of probabilities.

Chapter 2

Combinatorics for Probability

This chapter covers the combinatorial methods that are essential for calculating probabilities in a systematic way.

Chapter 3

Probability Distributions (Selected Models)

This chapter introduces several important probability distributions that are used to model various random phenomena.

Chapter 4

Discovering Formalism

This chapter focuses on the conceptual development of probability theory, moving from intuitive ideas to a more formal mathematical framework.

Chapter 5

PMF, PDF, and CDF

This chapter provides a technical introduction to the standard representations of probability distributions: the Probability Mass Function (PMF), the Probability Density Function (PDF), and the Cumulative Distribution Function (CDF).

Chapter 6

Algebra of Events and Conditional Probability

This chapter explores the algebra of events and the concept of conditional probability, which are fundamental to understanding how probabilities are updated with new information.

Chapter 7

Descriptive Statistics from Generated Data

This chapter shifts the perspective from theoretical probability models to the analysis of data using descriptive statistics.